

Research Notes

1 Introduction

Document will outline my approach to learning hydraulics and how to create a hydraulic launcher. Hydraulic launchers are interesting because they are used often in rollercoaster launch systems, see figure 1, military drones, and aircrafts. Often, large drones are catapulted with hydraulic launchers because of their ability to deliver high force, and precise speed in a compact area.



Figure 1: *Wicked* rollercoaster from theme park in Utah [1].

I'll detail in this report the exact mechanics of hydraulics and how one can fabricate one from scratch.

2 Hydraulics

References

- [1] Nick Weisenberger. Launch systems. *Coasters-101*, 2013.